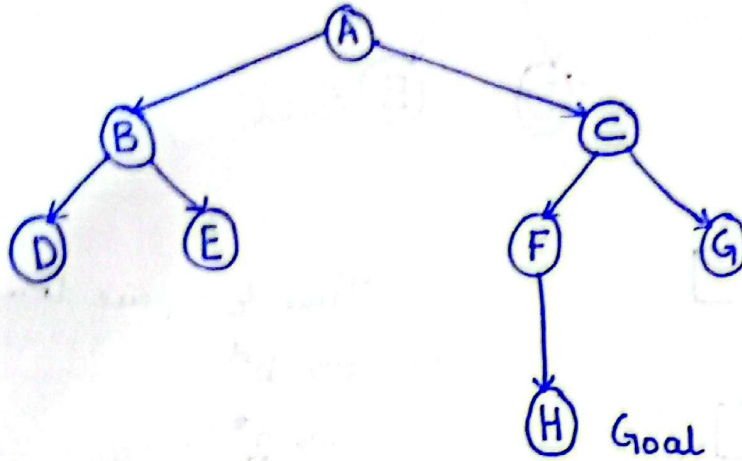
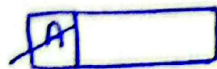


Q1. Breadth First Search:



Step 1:



Step 2:



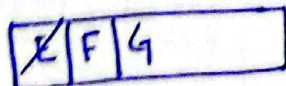
Step 3:



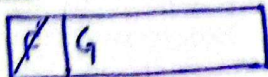
Step 4:



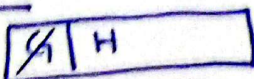
Step 5:



Step 6:



Step 7:



Step 8:



Goal Achieved!

Time & Space Complexity

$$\Rightarrow b^d$$

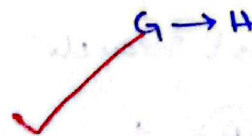
$$\Rightarrow 2^3$$

$$\Rightarrow 8$$

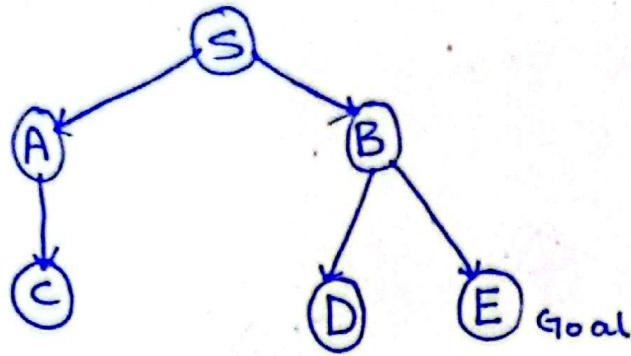
Path:

A → B → C → D → E → F →

G → H



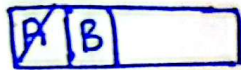
Q 2,



Step:1



Step:2



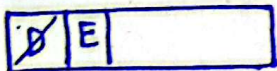
Step:3



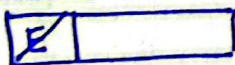
Step:4



Step:5



Step:6



Goal Achieved!

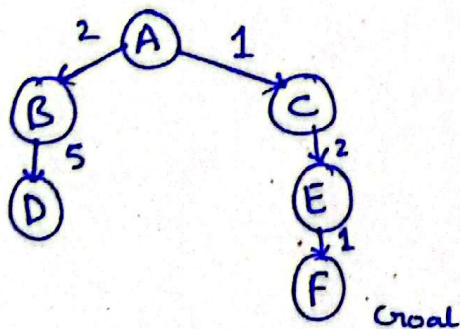
Time & Space complexity

$\Rightarrow b^d$

$\Rightarrow 2^2$

$\Rightarrow 4$

Q3, Uniform Cost Search



Step 1:

~~C(1), B(2)~~

Step 2:

~~B(2), E(3)~~

Step 3:

~~E(3), D(7)~~

Step 4:

~~F(4), D(7)~~

Goal Achieved!

Path:

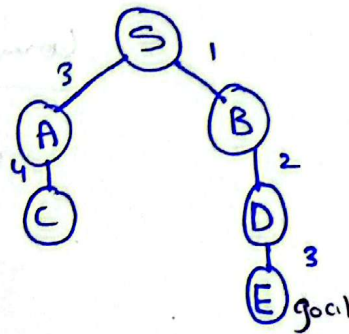
A → C → E → F

Time Complexity:

$$b^{c/e} = 2^{4/1} = 16$$

Cost: 4

Q4, UCF



Step 1:

~~B(1), A(3)~~

Step 2:

~~A(3), D(3)~~

Step 3:

~~D(3), C(7)~~

Step 4:

~~E(6), C(7)~~

Goal Achieved!

Path:

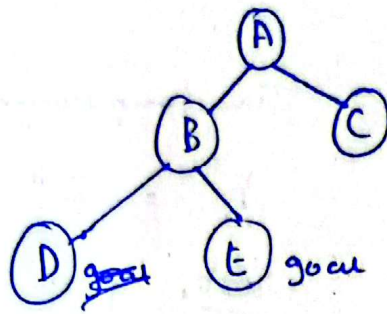
S → B → D → E

Time Complexity:

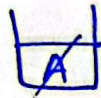
$$b^{c/e} = 2^{6/1} = 64$$

Cost: 6

Q5, Depth First Search



Step 1

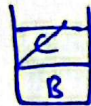


Path:

A → C → B → E



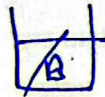
Step 2



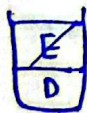
Complexity:

$$b^d = 2^2 = 4$$

Step 3

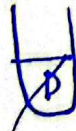


Step 4

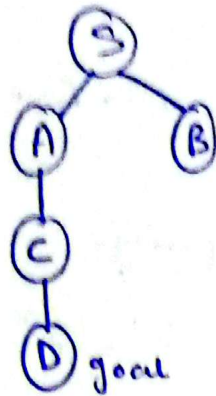


Goal Achieved!

Step 5



Q6, DFS



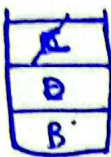
Step 1:



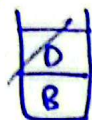
Step 2:



Step 3:



Step 4:



Goal Achieved!

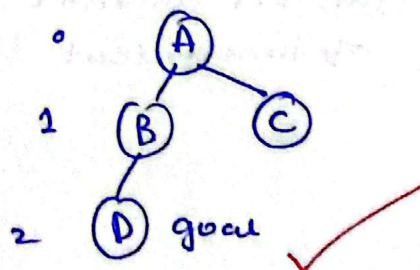
Path:

S → A → C → D

Complexity:

$$bd = 2^3 \Rightarrow 8$$

Q7, Depth Limited Search (L=2)



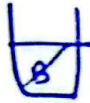
Step 1:



Step 2:



Step 3:



Step 4:



Goal Achieved! ✓

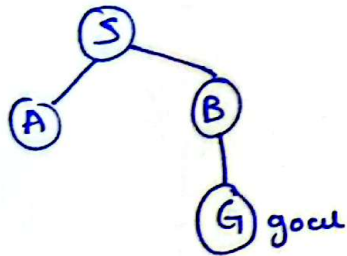
Path:

A → B → B → D

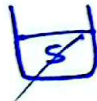
Complexity:

$$b^d = 2^2 = 4$$

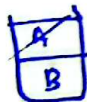
Q8, DLF (L=1)



Step 1:



Step 2:

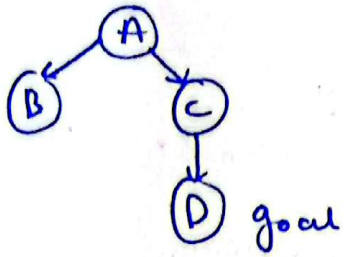


Step 3:



This DLF couldn't complete because of insufficient Unit. ✓

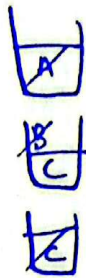
Q9, Iterative Deeping Search:



Iteration 1



Iteration 2



Iteration 3



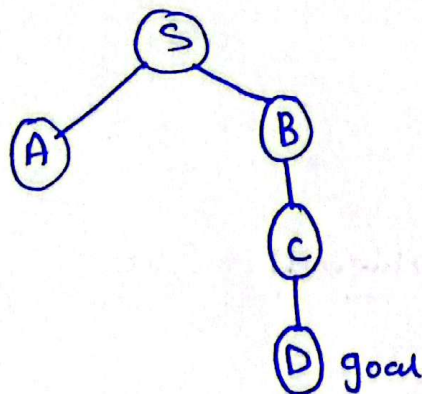
Path:

A → C → D ✓

Complexity:

$$b^d = 2^2 \Rightarrow 4$$

Q90, IDS



Iteration: 1

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Iteration: 2

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Iteration: 3

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Iteration: 4

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Goal Achieved!

Path:

$S \rightarrow B \rightarrow C \rightarrow D$

Complexity:

$ba \Rightarrow 2^3 \Rightarrow 8$